

**CMPE 462
MIDTERM EXAMINATION**

**Machine Problem: Data Cleaning and
Machine Learning**

**SVM Image Classification
of Healthy and Diseases Mango Leaf**

“Clean Data, Smart Models: From Preprocessing to Prediction”

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Objectives

Students should be able to:

- Perform **data cleaning and preprocessing**
 - Apply machine learning techniques:
 - **Regression (tabular data)** OR
 - **Image Classification using Support Vector Machine (SVM)**
 - Evaluate model performance
 - Interpret and explain results
-

PART 0

Identifying the Problem and Proposed Solution

Specific Real-World Problem

Mango diseases are often detected too late because many farmers rely only on manual visual inspection. In rural areas, access to agricultural experts is limited, causing delayed treatment and allowing infections to spread across plantations. This leads to:

- Reduced mango yield
- Lower fruit quality
- Financial losses for farmers
- Increased pesticide waste due to incorrect treatment

Specific Solution

The system uses Support Vector Machine (SVM) image classification to automatically identify whether a mango leaf is healthy or diseased through captured images. By providing faster initial detection, the system can help farmers:

- Isolate infected trees earlier
- Apply proper treatment sooner
- Reduce unnecessary pesticide usage
- Minimize disease spread in mango farms

Stronger Real-World Impact

This can specifically help:

- Small-scale mango farmers who cannot regularly access agricultural specialists
- Local farms needing low-cost crop monitoring
- Agricultural offices conducting early disease surveillance

Since mango is an important agricultural product in the Philippines, early disease detection can contribute to:

- Protecting crop productivity
- Reducing economic losses in farming communities
- Improving food and agricultural sustainability

PART 1

Data Cleaning

Dataset

Title	Link:
MangoLeafBD For Classification Of Mango Leaf Diseases	https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset

Data Exploration

Code:

```
[1]: import pandas as pd
import os
import numpy as np
import cv2

# Packages for visuals
import matplotlib.pyplot as plt

# Allows charts to appear in the notebook
%matplotlib inline

import pickle

[2]: dir = 'C:/Users/diyonah/DS2/G4 Midterm Project/Mango Leaf'
```

```
[3]: categories = ['Healthy','Not Healthy']
data = []

for category in categories:
    path = os.path.join(dir,category) #'C:/Users/diyonah/DS2/G4 Midterm Project/Mango Leaf'
    label = categories.index(category)
    for img in os.listdir(path):
        imgpath = os.path.join(path,img) #'C:/Users/diyonah/DS2/G4 Midterm Project/Mango Leaf'
        leaf_img = cv2.imread(imgpath,0)
        try:
            leaf_img = cv2.resize(leaf_img,(50,50))
            image = np.array(leaf_img).flatten()
            data.append([image,label])
        except Exception as e:
            pass
print(len(data))
```

200

[4]: data

```
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```

```
[5]: import pickle
```

```
pick_in = open('MANGOLEAF_DATASET.pickle','wb') #create a pickle file and set to write mode
pickle.dump(data,pick_in)
pick_in.close()
```

```
[6]: import pickle
```

```
pick_out = open('MANGOLEAF_DATASET.pickle','rb') #read a pickle file and set to write mode
mydata = pickle.load(pick_out)
pick_out.close()
```

```
[7]: mydata
```

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[array([213, 214, 218, ..., 239, 239, 239], shape=(2500,), dtype=uint8), 0],
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```

```
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```



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array([ 63, 63, 63, ..., 137, 131, 129], shape=(2500,), dtype=uint8), 1,
array([217, 212, 213, ..., 210, 211, 210], shape=(2500,), dtype=uint8), 1,
array([204, 203, 203, ..., 170, 170, 170], shape=(2500,), dtype=uint8), 1]
]
```

```
[8]: import random
```

```
random.shuffle(mydata)
features = []
labels = []

for feature, label in mydata:
    features.append(feature)
    labels.append(label)
```



```
[9]: from sklearn.model_selection import train_test_split
from sklearn.svm import SVC #support vector classifier

xtrain, xtest, ytrain, ytest = train_test_split(features, labels, test_size=0.20)
model = SVC(C=1, kernel='poly', gamma='auto')
model.fit(xtrain, ytrain)
```

PART 2

Image Classification using SVM

Model Implementation

Code:

```
[9]: from sklearn.model_selection import train_test_split
from sklearn.svm import SVC #support vector classifier

xtrain, xtest, ytrain, ytest = train_test_split(features, labels, test_size=0.20)
model = SVC(C=1, kernel='poly', gamma='auto')
model.fit(xtrain, ytrain)
```

[9]: SVC(C=1, gamma='auto', kernel='poly')

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.



SVC

?Documentation for SVCifitted

▼ Parameters

C	1
kernel	'poly'
degree	3
gamma	'auto'
coef0	0.0
shrinking	True
probability	False
tol	0.001
cache_size	200
class_weight	None
verbose	False
max_iter	-1
decision_function_shape	'ovr'
break_ties	False
random_state	None

Image Preprocessing & Evaluation Metrics

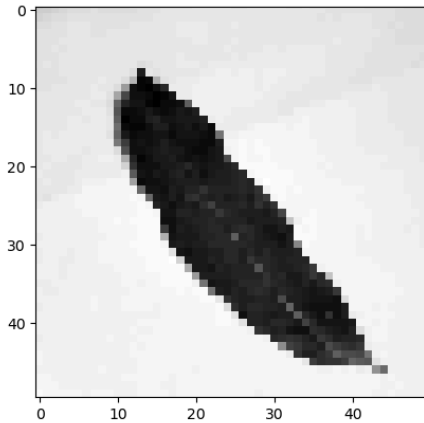
Code:

```
[10]: categories = ['Healthy','Not Healthy']
      prediction = model.predict(xtest)
      accuracy = model.score(xtest,ytest)

      print('Accuracy: ',accuracy)
      print('Prediction is: ',categories[prediction[0]])

      leaf = xtest[min(0, len(xtest)-1)].reshape(50,50)
      plt.imshow(leaf,cmap='gray')
      plt.show()
```

Accuracy: 0.825
Prediction is: Healthy



Visualization

Code:

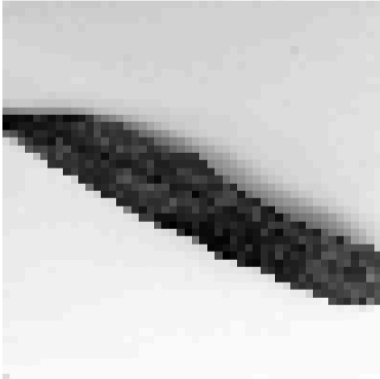
```
[11]: # Highlight misclassified images
      misclassified_indices = [i for i, (pred, true) in enumerate(zip(prediction, ytest)) if pred != true]
      print(f"Number of misclassified images: {len(misclassified_indices)}")
      for idx in misclassified_indices[:5]: # Show up to 5 misclassified samples
          img = np.array(xtest[idx]).reshape(50, 50)
          plt.imshow(img, cmap='gray')
          plt.title(f"True: {categories[ytest[idx]]}, Pred: {categories[prediction[idx]]}")
          plt.axis('off')
          plt.show()
```

Number of misclassified images: 7

True: Not Healthy, Pred: Healthy



True: Not Healthy, Pred: Healthy



True: Healthy, Pred: Not Healthy



True: Not Healthy, Pred: Healthy



True: Healthy, Pred: Not Healthy



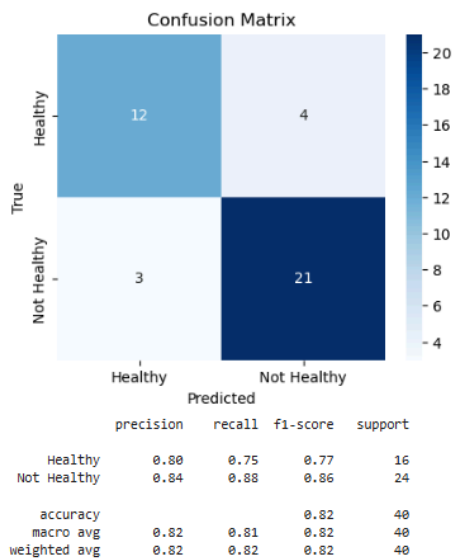
Classification Report

Code:

```
[12]: from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

# Confusion Matrix
cm = confusion_matrix(ytest, prediction)
plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=categories, yticklabels=categories)
plt.xlabel('Predicted')
plt.ylabel('True')
plt.title('Confusion Matrix')
plt.show()

# Classification Report
print(classification_report(ytest, prediction, target_names=categories))
```



```
[13]: len(prediction)
```

```
[13]: 40
```

PART 3

Testing

Mango Leaf 1:



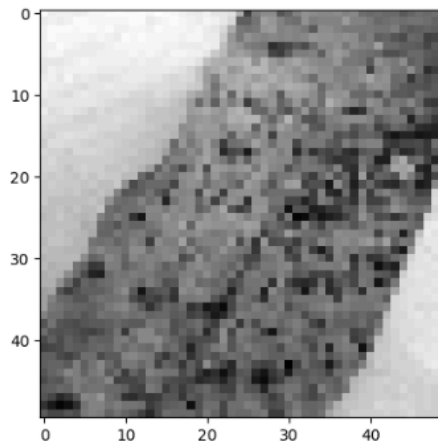
Testing:

```
[14]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (1).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 2:



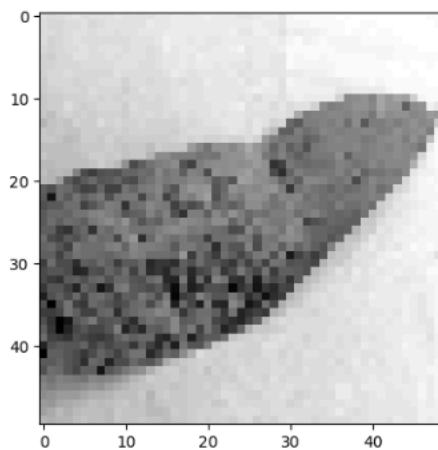
Testing:

```
[15]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (2).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 3:



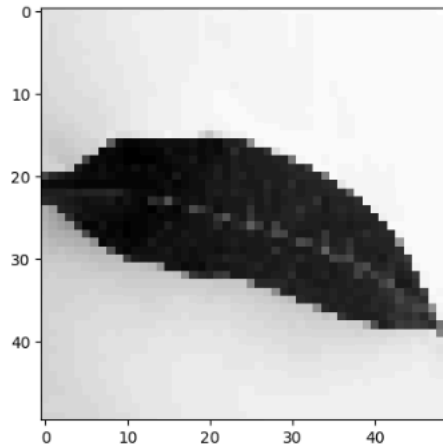
Testing:

```
[16]: categories = ['Healthy','Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (3).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 4:



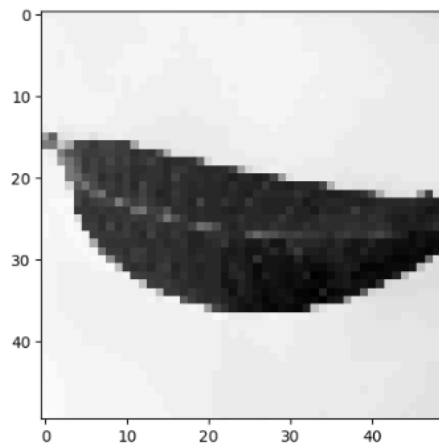
Testing:

```
[17]: categories = ['Healthy','Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (4).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 5:



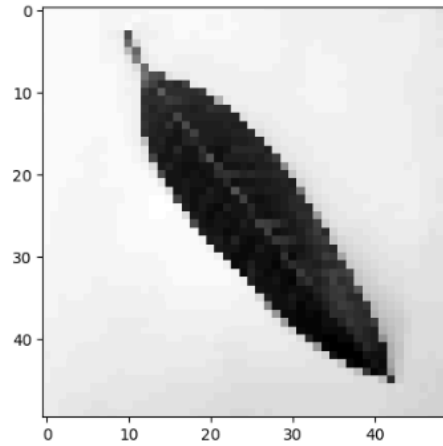
Testing:

```
[18]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (5).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 6:



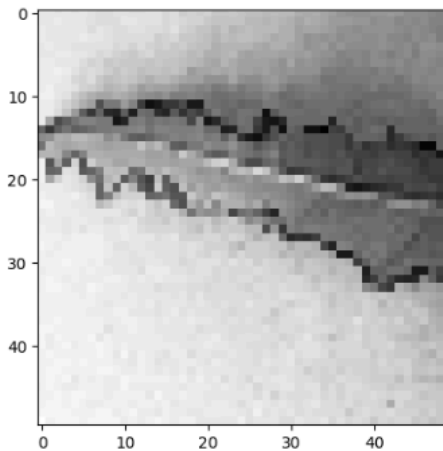
Testing:

```
[19]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (6).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 7:



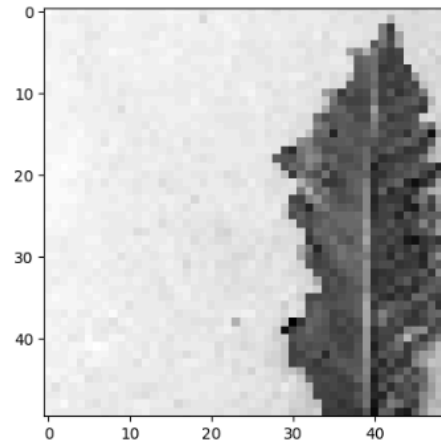
Testing:

```
[20]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (7).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 8:



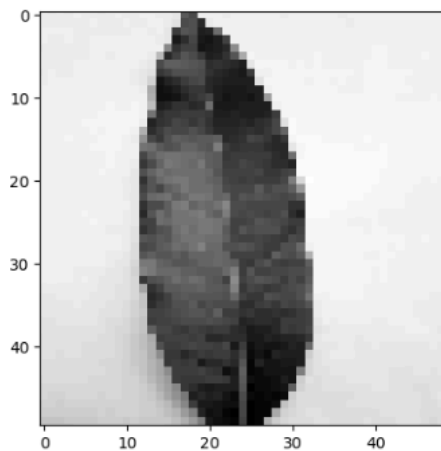
Testing:

```
[26]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (8).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 9:



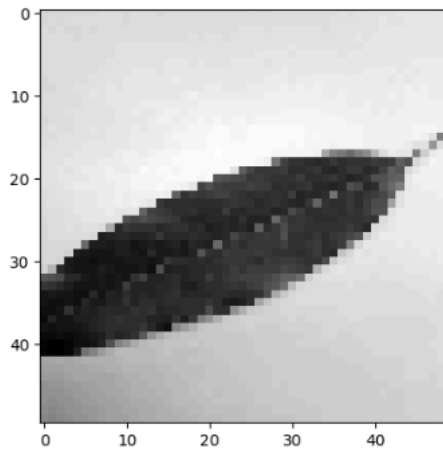
Testing:

```
[27]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (9).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 10:



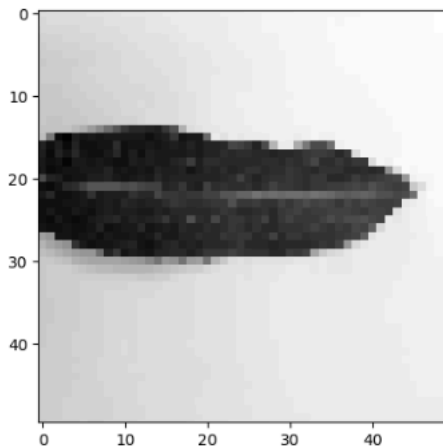
Testing:

```
[28]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (10).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 11:



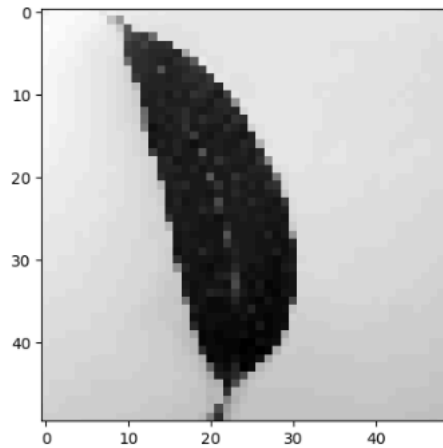
Testing:

```
[29]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (11).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 12:



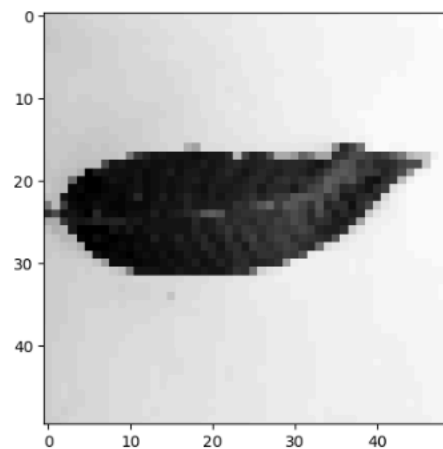
Testing:

```
[30]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (12).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 13:



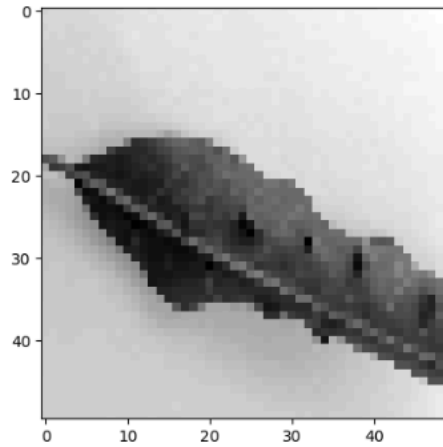
Testing:

```
[37]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (13).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 14:



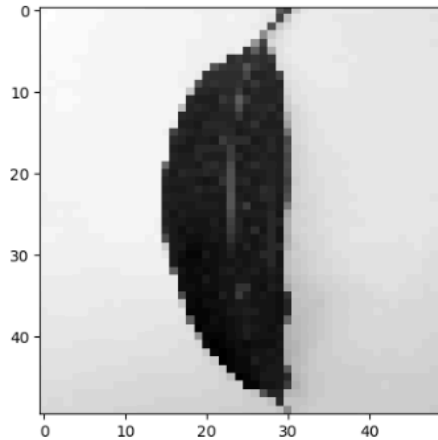
Testing:

```
[38]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (14).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 15:



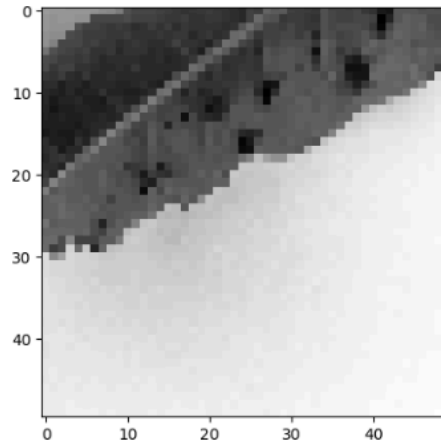
Testing:

```
[39]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (15).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 16:



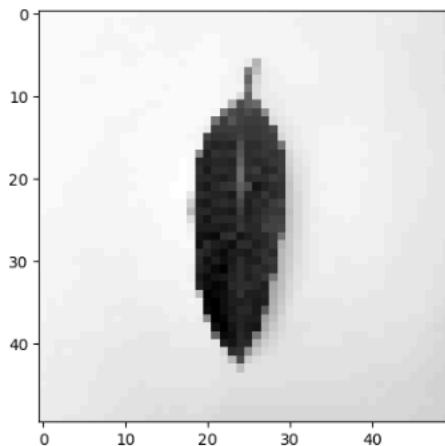
Testing:

```
[40]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (16).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Healthy



Mango Leaf 17:



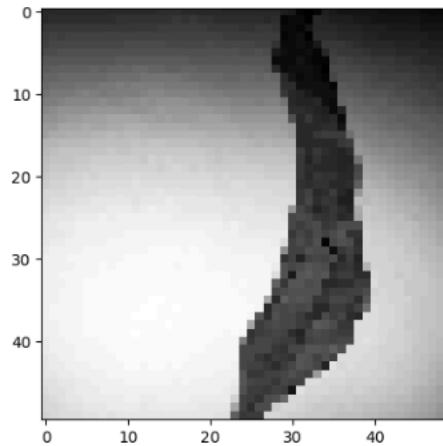
Testing:

```
[41]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (17).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 18:



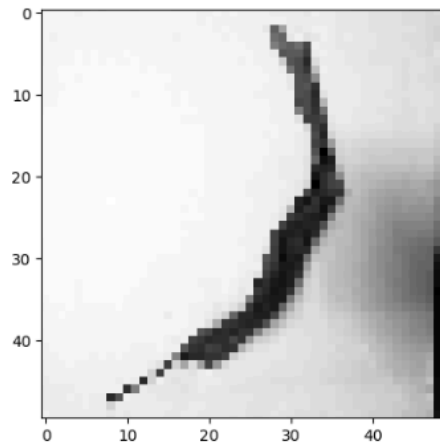
Testing:

```
[42]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (18).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 19:



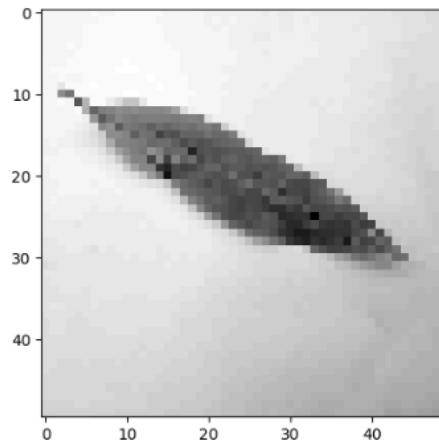
Testing:

```
[43]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (19).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Mango Leaf 20:



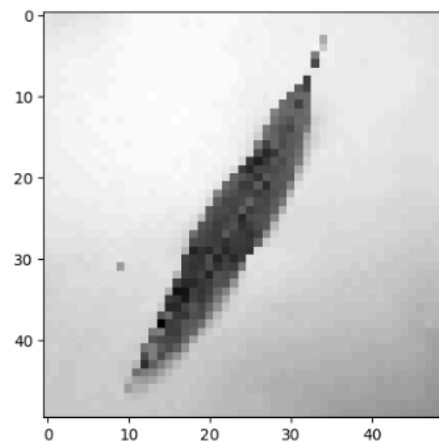
Testing:

```
[44]: categories = ['Healthy', 'Not Healthy']
leaf_img = cv2.imread('C:/Users/diyonnah/DS2/G4 Midterm Project/Mango Leaf/Test/Test (20).jpg',0)
leaf_img = cv2.resize(leaf_img,(50,50))
leaf_img = np.array(leaf_img).flatten()
leaf_img = leaf_img.reshape(1, -1)
pred = model.predict(leaf_img)

print('Prediction is: ',categories[pred[0]])

mango_leaf = leaf_img.reshape(50,50)
plt.imshow(mango_leaf,cmap='gray')
plt.show()
```

Prediction is: Not Healthy



Saving the Model

```
[45]: #saving the model  
pick = open('MANGO_LEAF_Classifier.sav','wb')  
pickle.dump(model,pick)  
pick.close()
```

```
[46]: #reading the model  
pick = open('MANGO_LEAF_Classifier.sav','rb')  
model = pickle.load(pick)  
pick.close()
```